MA1522 Linear Algebra for Computing
L2 Lesson 1



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Semester 1, 2026/27

Linear Equation

A **linear equation** in variables $x_1, x_2, \ldots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, a_2, \ldots, a_n$ and b are constants.

1. If $a_1 = a_2 = \cdots = a_n = 0 = b$, it is a **zero equation**.
   $(x_1, x_2, \ldots, x_n) = (t_1, t_2, \ldots, t_n)$ can take any values.
2. If $a_1 = a_2 = \cdots = a_n = 0 \neq b$, the equation is **inconsistent**.
   The linear equation has no solution.
3. Suppose that not all $a_1, a_2, \ldots, a_n$ are zero.
   - Fix any variable x_i such that $a_i \neq 0$.
   - Set the other $n-1$ variables as arbitrary parameters.
   - Solve x_i in terms of these $n-1$ arbitrary parameters.

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Question 1

a. Find a linear equation in variables x, y such that it has a general solution

$$x = 1 + 2t,\ y = t,$$

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The variables x and y are both expressed in parameter t.

The parameter t can be written in terms of x, and in terms of y:

$$t = \frac{1}{2}x - \frac{1}{2} \text{ and } t = y.$$

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Thus $\frac{1}{2}x - \frac{1}{2} = y \ \Rightarrow\ \frac{1}{2}x - y = \frac{1}{2}$, or $x - 2y = 1$. □

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(b) Show that $x = t$ and $y = \frac{1}{2}t - \frac{1}{2}$, $t \in \mathbb{R}$, is also a general solution to the equation obtained in (a).

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Solution.

Set $x = t$ as an arbitrary parameter. Then

$$t - 2y = 1 \;\Rightarrow\; 2y = t - 1 \;\Rightarrow\; y = \frac{1}{2}t - \frac{1}{2}.$$



Question 2

Find a linear equation in variables x, y, z which has a general solution

$$\begin{cases} x = 3 - 4s + t, \\ y = s, \\ z = t, \end{cases}$$

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Solution.

The values of $y = s$ and $z = t$ are arbitrary parameters. Substitute them into the first equation:

$$x = 3 - 4s + t = 3 - 4y + z.$$

The linear equation: $x + 4y - z = 3$. □

Linear System

A **linear system** in variables $x_1, x_2, \ldots, x_n$ has the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

- a_{ij} is the coefficient in the ith equation of the jth variable.
- b_i is the constant of the ith equation.

Its associated **augmented matrix** is

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right).$$

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Elementary Row Operations

The following are the **elementary row operations** for a matrix:

- Interchange two rows: $R_i \leftrightarrow R_j$ $(i \neq j)$.
- Add a nonzero multiple c of a row to another row: $R_i + cR_j$ $(i \neq j)$.
- Multiply a row by a nonzero constant a: aR_i $(a \neq 0)$.

Matrices A and B are **row equivalent** if one is obtained from the other by a **series** of **elementary row operations**:

- $A = A_0 \to A_1 \to A_2 \to \cdots \to A_{k-1} \to A_k = B$, or
- $A = A_0 \leftarrow A_1 \leftarrow A_2 \leftarrow \cdots \leftarrow A_{k-1} \leftarrow A_k = B$.

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Row-echelon and Reduced Row-echelon Forms

A matrix is said to be in **row-echelon form** if:

1. the zero rows (if any) are grouped together at the bottom of the matrix;
2. the first nonzero entry (called the **leading entry**) of each nonzero row moves to the right as we go down the rows.

A matrix is said to be in **reduced** row-echelon form if:

1. it is in row-echelon form;
2. the leading entry of every nonzero row is 1;
3. all other entries of a column containing a leading entry (such a column is called a **pivot column**) are 0.

A **(reduced) row-echelon form of a matrix A** is a matrix that is

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Gaussian Elimination

Let $\boldsymbol{A}$ be a matrix. **Gaussian elimination** is an algorithm to obtain **a** row-echelon form of $\boldsymbol{A}$, as follows:

1. Find the leftmost nonzero column of $\boldsymbol{A}$, say column j.
2. If the top entry a_{1j} of such column of $\boldsymbol{A}$ is zero, swap the top row with row i which has $a_{ij} \neq 0$.
3. For each row $i \geq 2$, add a suitable multiple of the top row to it so that the resultant (i, j)-entry equals 0.
4. Cover the top row and repeat (1)–(3) to the remaining matrix.
5. Repeat (4) until the matrix is in row-echelon form.

Note

Gaussian elimination only involves elementary row operations of the form $R_i \leftrightarrow R_j$ and $R_k + cR_l$ with $k > l$.

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Gauss-Jordan Elimination

Let $\boldsymbol{A}$ be a matrix. **Gauss-Jordan elimination** is an algorithm to obtain **the** reduced row-echelon form of $\boldsymbol{A}$, as follows:

1. Use Gaussian elimination to obtain a row-echelon form of $\boldsymbol{A}$.
2. Multiply each nonzero row with a suitable constant so that its leading entry equals 1.
3. Beginning with the bottommost nonzero row and working upwards, add a suitable multiple of this row to the rows above it to obtain 0's directly above its leading entry.

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Solving Linear System

1. Write down the augmented matrix $\boldsymbol{A}$ of the given linear system.
2. Apply Gaussian elimination or Gauss-Jordan elimination to A.
   - Gaussian elimination: obtain row-echelon form R.
   - Gauss-Jordan elimination: obtain reduced row-echelon form R.
3. Solve the linear system of R.
   a. Set variables corresponding to **non-pivot columns** of R as arbitrary parameters.
   b. Solve variables corresponding to **pivot columns** of R.
      - Use back substitution if R is not in reduced row-echelon form.
      - Use any order if R is in the reduced row-echelon form.

Remarks

The system is **consistent** if and only if the rightmost column (corresponding to constants) of R is **non-pivot**.

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   - Gaussian elimination: obtain row-echelon form $\boldsymbol{R}$.
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Question 3

Solve the following linear system: $\begin{cases} 3x_1 + 2x_2 - 4x_3 = 3 \\ 2x_1 + 3x_2 + 3x_3 = 15 \\ 5x_1 - 3x_2 + x_3 = 14 \end{cases}$

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Solution.

$$\left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 2 & 3 & 3 & 15 \\ 5 & -3 & 1 & 14 \end{array}\right) \xrightarrow[R_3 - \frac{5}{3}R_1]{R_2 - \frac{2}{3}R_1} \left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 0 & \frac{5}{3} & \frac{17}{3} & 13 \\ 0 & -\frac{19}{3} & \frac{23}{3} & 9 \end{array}\right)$$

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Solution.

$$\left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 2 & 3 & 3 & 15 \\ 5 & -3 & 1 & 14 \end{array}\right) \xrightarrow[R_3 - \frac{5}{3}R_1]{R_2 - \frac{2}{3}R_1} \left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 0 & \frac{5}{3} & \frac{17}{3} & 13 \\ 0 & -\frac{19}{3} & \frac{23}{3} & 9 \end{array}\right)$$

$$\xrightarrow{R_3 + \frac{19}{5}R_2} \left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 0 & \frac{5}{3} & \frac{17}{3} & 13 \\ 0 & 0 & \frac{146}{5} & \frac{292}{5} \end{array}\right) \xrightarrow[\frac{5}{146}R_3]{\frac{1}{3}R_1 \\ \frac{3}{5}R_2} \left(\begin{array}{ccc|c} 1 & \frac{2}{3} & -\frac{4}{3} & 1 \\ 0 & 1 & \frac{17}{5} & \frac{39}{5} \\ 0 & 0 & 1 & 2 \end{array}\right)$$

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Solve the following linear system: $\begin{cases} 3x_1 + 2x_2 - 4x_3 = 3 \\ 2x_1 + 3x_2 + 3x_3 = 15 \\ 5x_1 - 3x_2 + x_3 = 14 \end{cases}$

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$$\xrightarrow[R_2 - \frac{17}{5}R_3]{R_1 + \frac{4}{3}R_3} \left(\begin{array}{ccc|c} 1 & \frac{2}{3} & 0 & \frac{11}{3} \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{R_1 - \frac{2}{3}R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

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Solution (cont'd).

- $\left(\begin{array}{ccc|c} 3 & 2 & -4 & 3 \\ 2 & 3 & 3 & 15 \\ 5 & -3 & 1 & 14 \end{array}\right) \xrightarrow[\text{Jordan}]{\text{Gauss-}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right)$
- The rightmost column is non-pivot. Therefore the system is consistent.
- The columns corresponding to x_1, x_2, x_3 are pivot. So there is no arbitrary parameter.
- Solve variables x_1, x_2, x_3:

$$x_1 = 3, \quad x_2 = 1, \quad x_3 = 2.$$

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Question 4

Solve the following linear system:

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Solution.

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 0 \\ 2 & 1 & -1 & 1 & -2 \\ -1 & 1 & -3 & 1 & 4 \end{array}\right) \xrightarrow[R_3+R_1]{R_2-2R_1} \left(\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 0 \\ 0 & -1 & 1 & 5 & -2 \\ 0 & 2 & -4 & -1 & 4 \end{array}\right)$$

$$\xrightarrow{R_3+2R_2} \left(\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 0 \\ 0 & -1 & 1 & 5 & -2 \\ 0 & 0 & -2 & 9 & 0 \end{array}\right) \xrightarrow[-\frac{1}{2}R_3]{-R_2} \left(\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 0 \\ 0 & 1 & -1 & -5 & 2 \\ 0 & 0 & 1 & -\frac{9}{2} & 0 \end{array}\right)$$

$$\xrightarrow[R_2+R_3]{R_1+R_3} \left(\begin{array}{cccc|c} 1 & 1 & 0 & -\frac{13}{2} & 0 \\ 0 & 1 & 0 & -\frac{19}{2} & 2 \\ 0 & 0 & 1 & -\frac{9}{2} & 0 \end{array}\right) \xrightarrow{R_1-R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & -2 \\ 0 & 1 & 0 & -\frac{19}{2} & 2 \\ 0 & 0 & 1 & -\frac{9}{2} & 0 \end{array}\right).$$

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- The rightmost column is non-pivot. Therefore the system is consistent.
- The columns corresponding to variables a, b, c are pivot.
  - Set $d = t$ as an arbitrary parameter.
  - Solve variables a, b, c in terms of t:
    $$a = -3t - 2, \quad b = \frac{19}{2}t + 2, \quad c = \frac{9}{2}t.$$
  - Final solution:
    $$a = -3t - 2, \quad b = \frac{19}{2}t + 2, \quad c = \frac{9}{2}t, \quad d = t, \quad t \in \mathbb{R}.$$

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Next Lesson ...

We will continue exploring **Chapter 1**.